

Lecture 1: Difference Equation

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1. Difference operators

$$\text{1-st order difference: } \Delta y_t = y_t - y_{t-1}$$

$$\begin{aligned} \text{2-nd order difference: } \Delta^2 y_t &= \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = \Delta y_t - \Delta y_{t-1} = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) \\ &= y_t - 2y_{t-1} + y_{t-2}. \end{aligned}$$

2. Period

Period: an interval of time $\Rightarrow y_t$ is the y for t^{th} interval of time, not for a point of time.

* t is a period concept, not a point-in-time concept.

$$\Delta y_t = y_t - y_{t-1}$$

is the change from $(t-1)^{\text{th}}$ time interval to t^{th} interval, not a derivative at a point of time t .

* Δy_t is a change, not a derivative or growth rate at time t .

$$\text{Derivative} = \dot{y}_t = \frac{\Delta y_t}{y_{t-1}} = \frac{y_t - y_{t-1}}{y_{t-1}} = \frac{y_t}{y_{t-1}} - 1 \approx \log\left(\frac{y_t}{y_{t-1}}\right)$$

$$\text{when } \frac{y_t - y_{t-1}}{y_{t-1}} \text{ is small, } \log\left(1 + \frac{y_t}{y_{t-1}} - 1\right) \approx \frac{y_t}{y_{t-1}} - 1 = \frac{y_t - y_{t-1}}{y_{t-1}}.$$

3. Solve a first-order difference equation

Solve a first-order difference equation $\iff$ rewrite y_t as a function of t .

Example:

$$y_t + ay_{t-1} = C.$$

(1) Brutal force: iterative method

$$y_t = -ay_{t-1} + C$$

$$y_{t-1} = -ay_{t-2} + C$$

Hence

$$y_t = a^2 y_{t-2} - aC + C, \quad y_{t-2} = -a y_{t-3} + C.$$

Continuing,

$$y_t = -a^3 y_{t-3} + a^2 C - aC + C - \dots$$

and in general,

$$y_t = (-a)^t y_0 + C \sum_{j=0}^{t-1} (-a)^j.$$

If $a \neq -1$,

$$\sum_{j=0}^{t-1} (-a)^j = \frac{1 - (-a)^t}{1 + a}.$$

So

$$y_t = (-a)^t \left(y_0 - \frac{C}{1+a} \right) + \frac{C}{1+a}, \quad a \neq -1.$$

Equivalently,

$$y_t = A(-a)^t + \frac{C}{1+a}, \quad a \neq -1.$$

If $a = -1$,

$$\sum_{j=0}^{t-1} (-a)^j = \sum_{j=0}^{t-1} 1 = t \quad \Rightarrow \quad y_t = y_0 + Ct.$$

Thus

$$y_t = \begin{cases} A(-a)^t + \frac{C}{1+a}, & a \neq -1, \\ Ct + y_0, & a = -1, \end{cases} \quad \text{where } A = y_0 - \frac{C}{1+a}.$$

(2) General method

Since the system is *linear*, we can separate:

- 1) the part driven by constant C ,
- 2) the part comes from the dynamics.

Why $y_t = y_t^p + y_t^c$ is valid?

Define linear operator

$$L(\cdot) = (\cdot)_t + a(\cdot)_{t-1}.$$

Then

$$y_t + a y_{t-1} = C \quad \Rightarrow \quad L(y) = C.$$

If y^p is any particular solution such that

$$L(y^p) = C,$$

and if y^c is any solution of the homogeneous equation

$$L(y^c) = 0 \quad (\text{complementary solution}),$$

then by linearity,

$$L(y^p + y^c) = L(y^p) + L(y^c) = C + 0 = C.$$

Therefore,

$$y^p + y^c$$

solves the equation

$$y_t + ay_{t-1} = C.$$

All solutions = one particular solution + all homogeneous solutions.

Now we solve the difference equation with general method.

1) Complementary solution Since exponential sequences are eigenfunctions of the lag operator, try

$$y_t = Ab^t.$$

Then for the homogeneous equation,

$$Ab^t + aAb^{t-1} = 0 \Rightarrow b + a = 0 \Rightarrow b = -a.$$

Hence

$$y_t^c = A(-a)^t.$$

2) Particular solution Since the forcing term is a constant C , we guess

$$y_t^p = k.$$

Then

$$k + ak = C \Rightarrow k = \frac{C}{1+a}, \quad a \neq -1.$$

If $a = -1$,

$$y_t - y_{t-1} = C \Rightarrow y_t^p = Ct.$$

Therefore,

$$y_t = y_t^c + y_t^p = \begin{cases} A(-a)^t + \frac{C}{1+a}, & a \neq -1, \\ A + Ct, & a = -1. \end{cases}$$

Using the initial condition:

If $a \neq -1$,

$$y_0 = A(-a)^0 + \frac{C}{1+a} = A + \frac{C}{1+a} \Rightarrow A = y_0 - \frac{C}{1+a}.$$

If $a = -1$,

$$y_0 = A \Rightarrow A = y_0.$$

4. Dynamic stability of equilibrium

Consider

$$y_t + ay_{t-1} = C.$$

The steady state is defined by

$$y_t = y_{t-1} = \bar{y},$$

so

$$\bar{y} + a\bar{y} = C \Rightarrow \bar{y} = \frac{C}{1+a} \quad (a \neq -1).$$

Subtract steady state from the original equation:

$$y_t - \bar{y} + a(y_{t-1} - \bar{y}) = C - \bar{y} - a\bar{y}.$$

Define deviation from steady state as

$$\tilde{y}_t = y_t - \bar{y}.$$

Then

$$\tilde{y}_t + a\tilde{y}_{t-1} = 0 \Rightarrow \tilde{y}_t = (-a)\tilde{y}_{t-1}.$$

Let

$$b = -a.$$

Then

$$\tilde{y}_t = b\tilde{y}_{t-1} \Rightarrow \tilde{y}_t = b^t \tilde{y}_0.$$

* $b = -a$ controls convergence.

$$\begin{cases} |b| < 1 \Rightarrow b^t \rightarrow 0, \tilde{y}_t \rightarrow 0, y_t \rightarrow \bar{y} & (\text{convergence}), \\ |b| > 1 \Rightarrow b^t \rightarrow \infty, \tilde{y}_t \rightarrow \infty & (\text{divergence}). \end{cases}$$

5. Sign of b and oscillation / monotonicity

Sign of b also controls **oscillation** / **monotonicity**.

$$\begin{cases} b > 0 : & b^t \text{ keeps the same sign for all } t \Rightarrow \tilde{y}_t \text{ keeps the same sign for all } t, \\ & \text{path is non-oscillatory (monotone, towards / away from } \bar{y} \text{).} \\ b < 0 : & b^t \text{ alternates sign} \Rightarrow \tilde{y}_t \text{ flips sign each period} \\ & \Rightarrow \text{oscillate around } \bar{y}. \end{cases}$$

Further:

$$\begin{cases} b > 1, & y_t \text{ monotonically diverges from } \bar{y}, \\ b = 1, & y_t \text{ keeps the same distance away from } \bar{y}, \\ 0 < b < 1, & y_t \text{ monotonically converge to } \bar{y}, \\ b = 0, & y_t \text{ stays at steady state } \bar{y} \text{ always,} \\ -1 < b < 0, & y_t \text{ oscillate around and converge to } \bar{y}, \\ b = -1, & y_t \text{ oscillate around } \bar{y} \text{ and keeps same absolute distance,} \\ b < -1, & y_t \text{ oscillate around and diverge from } \bar{y}. \end{cases}$$

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6. The role of A

Since the general solution can be written as

$$y_t = \bar{y} + Ab^t \Rightarrow A = y_0 - \bar{y}$$

(initial deviation).

Scale effect:

$|A| \uparrow \Rightarrow$ start farther from equilibrium $\Rightarrow$ bigger movements along path.

Mirror effect:

$A > 0$: start above $\bar{y}$; $A < 0$: start below $\bar{y}$.

7. Example: Inventory model

Example: Inventory model.

$$Q_{d,t} = \alpha - \beta p_t \quad \text{demand}$$

$$Q_{s,t} = -\gamma + \delta p_t \quad \text{supply}$$

$$p_{t+1} = p_t - \sigma (Q_{s,t} - Q_{d,t})$$

$$\alpha, \beta, \gamma, \delta, \sigma > 0 \quad \text{solve for } p_t \text{'s path}$$

Then

$$Q_{s,t} - Q_{d,t} = -(\alpha + \gamma) + (\beta + \delta)p_t.$$

Plug in:

$$p_{t+1} = [1 - \sigma(\beta + \delta)]p_t + \sigma(\alpha + \gamma).$$

At steady state,

$$p_{t+1} = p_t = \bar{p} \Rightarrow \bar{p} = \frac{\alpha + \gamma}{\beta + \delta}.$$

Thus, the deviation dynamics is

$$p_t - \bar{p} = [1 - \sigma(\beta + \delta)](p_{t-1} - \bar{p}).$$

Stability condition tells us:

$$|1 - \sigma(\beta + \delta)| < 1 \Rightarrow 0 < \sigma(\beta + \delta) < 2,$$

system converges.

Oscillation condition tells us:

$$1 - \sigma(\beta + \delta) < 0 \Rightarrow \sigma(\beta + \delta) > 1$$

creates oscillation.

* **Economic intuition:** if price adjustment is too aggressive $\sigma \uparrow\uparrow$, creates oscillation and divergence.

If supply or demand elasticity in terms of price too high

$$\beta \uparrow\uparrow / \delta \uparrow\uparrow$$

$\Rightarrow$ creates oscillation or even divergence.

8. Higher-order difference equations

2-nd order linear difference equation with constant coefficients & constant term:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = C.$$

1° Particular solution y^p : the intertemporal equilibrium level

Try a constant:

$$y^p = k \Rightarrow k(1 + a_1 + a_2) = C \Rightarrow k = \frac{C}{1 + a_1 + a_2}, \quad 1 + a_1 + a_2 \neq 0.$$

Try a first-degree polynomial:

$$y^p = kt.$$

Then

$$\begin{aligned} k(t+2) + a_1 k(t+1) + a_2 kt &= C \\ \Rightarrow k[(1 + a_1 + a_2)t + (2 + a_1)] &= C. \end{aligned}$$

Impose

$$1 + a_1 + a_2 = 0$$

then

$$k = \frac{C}{2 + a_1}, \quad 2 + a_1 \neq 0.$$

Try a second-degree polynomial:

$$y^p = kt^2.$$

Then

$$k(t+2)^2 + a_1 k(t+1)^2 + a_2 kt^2 = C.$$

The notes impose

$$\begin{cases} 1 + a_1 + a_2 = 0, \\ 2 + a_1 = 0, \end{cases} \Rightarrow \begin{cases} a_1 = -2, \\ a_2 = 1. \end{cases}$$

Substituting:

$$\begin{aligned} k(t^2 + 4t + 4) - 2k(t^2 + 2t + 1) + kt^2 &= C \\ \Rightarrow 2k &= C \Rightarrow k = \frac{C}{2}. \end{aligned}$$

Therefore

$$y^p = \begin{cases} \frac{C}{1 + a_1 + a_2}, & 1 + a_1 + a_2 \neq 0, \\ \frac{C}{2 + a_1} t, & 1 + a_1 + a_2 = 0, a_1 \neq -2, \\ \frac{C}{2} t^2, & a_1 = -2, a_2 = 1. \end{cases}$$

2° Complementary solution y^c : deviation from the equilibrium in each period

Look at the homogeneous equation:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = 0.$$

Try

$$y_t = Ab^t.$$

Then

$$b^2 + a_1 b + a_2 = 0.$$

Case 1: Two distinct real roots If

$$a_1^2 - 4a_2 > 0,$$

then

$$y^c = A_1 b_1^t + A_2 b_2^t$$

where

$$b_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \quad b_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}.$$

Case 2: Repeated real root If

$$a_1^2 - 4a_2 = 0,$$

then

$$b = b_1 = b_2 = -\frac{a_1}{2},$$

and

$$y^c = A_3 b^t + A_4 t b^t.$$

Case 3: Complex roots If

$$a_1^2 - 4a_2 < 0,$$

then

$$b_{1,2} = h \pm vi, \quad h = -\frac{a_1}{2}, \quad v = \frac{\sqrt{4a_2 - a_1^2}}{2}.$$

So

$$y^c = A_5(h + vi)^t + A_6(h - vi)^t.$$

Using De Moivre's Theorem:

$$(h \pm vi)^t = R^t (\cos(\theta t) \pm i \sin(\theta t))$$

where

$$R = \sqrt{h^2 + v^2} = \sqrt{a_2}, \quad \cos \theta = \frac{h}{R} = -\frac{a_1}{2\sqrt{a_2}}, \quad \sin \theta = \frac{v}{R} = \sqrt{1 - \frac{a_1^2}{4a_2}}.$$

Hence

$$y^c = R^t (A_5 \cos \theta t + A_6 \sin \theta t),$$

where

$$R = \sqrt{a_2}, \quad \theta \text{ is defined as } \begin{cases} \cos \theta = -\frac{a_1}{2\sqrt{a_2}}, \\ \sin \theta = \sqrt{1 - \frac{a_1^2}{4a_2}}. \end{cases}$$

9. Convergence under higher-order difference equation

1° Distinct root: b_{dom} is the root with highest absolute value. Convergence requires

$$|b_{\text{dom}}| < 1.$$

2° Repeated root:

$$|b| < 1 \Rightarrow \text{converges.}$$

3° Complex root:

$$R > 1 : \text{explosive oscillations,} \quad R < 1 : \text{damped oscillations.}$$

10. Simultaneous difference equation (state-space form)

Let

$$x_t = y_{t+1}.$$

Then

$$\begin{cases} x_{t+1} + a_1 x_t + a_2 y_t = C, \\ x_t = y_{t+1}. \end{cases}$$

Therefore,

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Generalize:

$$X_{t+1} = AX_t + B$$

where

$$X_t = (x_{1t}, x_{2t}, \dots, x_{nt})$$

is state space.

A is a constant matrix with a_{ij} as elements ($i, j = 1, \dots, n$),

B is an array with B_i , $i = 1, \dots, n$ as elements.

In this example,

$$X_{t+1} = \begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix}, \quad A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Everything about dynamics and stability is controlled by eigenvalues of A .

Ignore constant term B to study deviation from steady state:

$$X_{t+1} = AX_t.$$

Try solution:

$$X_t = vb^t.$$

Plug in:

$$vb^{t+1} = A(vb^t) \Rightarrow bv = Av.$$

Here b is an eigenvalue of A , with eigenvector v .

A nonzero v exists iff

$$\det(A - bI) = 0.$$

11. Characteristic equation

Characteristic equation:

$$p(b) = |A - bI| = 0.$$

For a 2×2 matrix:

$$p(b) = b^2 - Tb + D = 0,$$

where

$$T = \text{trace}(A), \quad D = \det(A).$$

Also,

$$T = b_1 + b_2, \quad D = b_1 b_2,$$

so

$$p(b) = b^2 - (b_1 + b_2)b + b_1 b_2 = (b - b_1)(b - b_2).$$

Eigenvalues are the solution of the characteristic equation: b_1, b_2 .

Back to the example, since

$$A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix},$$

we have

$$T = \text{tr}(A) = -a_1, \quad D = \det(A) = a_2.$$

Thus characteristic equation:

$$b^2 + a_1 b + a_2 = 0.$$

Eigenvalues:

$$\frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}, \quad \text{if } a_1^2 > 4a_2.$$

12. Stability triangle on the (T, D) plane

1°

$$\Delta = T^2 - 4D.$$

$$\begin{cases} \Delta < 0 : & \text{complex roots} \Rightarrow \text{above red curve,} \\ \Delta > 0 : & \text{real roots} \Rightarrow \text{below red curve.} \end{cases}$$

2°

$$p(1) = 0, \quad p(-1) = 0$$

are the black curves.

$$p(1) = 1 - T + D, \quad \text{right to } p(1) = 0 \Rightarrow p(1) < 0.$$

$$p(-1) = 1 + T + D, \quad \text{left to } p(-1) = 0 \Rightarrow p(-1) < 0.$$

3°

$$D = 1 \text{ and } |b| = 1 \text{ for complex roots}$$

is the green curve.

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Why D controls $|b|$ in complex case?

If roots are complex conjugates

$$b_{1,2} = re^{\pm i\theta} \Rightarrow D = b_1 b_2 = r^2 \Rightarrow r = \sqrt{D}.$$

Stable ($r < 1$) implies

$$D < 1.$$

Thus, above green curve $|b| > 1$, below green curve $|b| < 1$.

*** Conditions for two stable roots (both inside unit circle)**

$$\begin{aligned} D < 1 \quad \text{and} \quad p(1) > 0 \quad \text{and} \quad p(-1) > 0 \\ \Rightarrow D < 1, \quad 1 - T + D > 0, \quad 1 + T + D > 0 \\ \Rightarrow D < 1 \quad \text{and} \quad |T| < 1 + D. \end{aligned}$$

$$\begin{cases} \text{complex roots with modulus} < 1, & \text{if } \Delta < 0 \ (T^2 < 4D), \\ \text{real roots both between } -1 \text{ and } 1, & \text{if } \Delta > 0 \ (T^2 > 4D). \end{cases}$$

*** Conditions for one stable root and one unstable root.**

$$p(1) < 0 \ \& \ p(-1) > 0 \quad \text{or} \quad p(1) > 0 \ \& \ p(-1) < 0.$$

That is,

$$\begin{cases} 1 - T + D < 0, \\ 1 + T + D > 0, \end{cases} \quad \text{or} \quad \begin{cases} 1 - T + D > 0, \\ 1 + T + D < 0. \end{cases}$$

Thus,

$$|T| > |1 + D|.$$

$p(1) \ \& \ p(-1)$ has opposite signs $\Rightarrow$ There is a root in the interval $(-1, 1)$.

Since

$$p(1) < 0, \ p(b) \rightarrow +\infty \text{ as } b \rightarrow +\infty$$

or

$$p(-1) < 0, \ p(b) \rightarrow +\infty \text{ as } b \rightarrow -\infty,$$

there is another root outside the interval $[-1, 1]$.

*** Conditions for explosive and unstable roots (both roots outside unit circle or complex with modulus > 1)**

Four cases:

$$1^\circ \ p(1) < 0, \ p(-1) < 0, \ D < 0, \ |T| > 1$$

$$2^\circ \ p(1) > 0, \ p(-1) > 0, \ D > 1, \ T < -2$$

$$3^\circ \ \Delta < 0 \text{ and } D > 1$$

$$4^\circ \ p(1) > 0, \ p(-1) > 0, \ D > 1, \ T > 2$$

13. Region-by-region summary

Region	Conditions	Roots
1	$p(1) < 0 \ \& \ p(-1) > 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1, 1), \\ 1 \text{ unstable root in } (1, +\infty) \end{cases}$
2	$p(1) < 0 \ \& \ p(-1) < 0$	explosive: $\begin{cases} 1 \text{ unstable root in } (1, +\infty), \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
3	$p(1) > 0 \ \& \ p(-1) < 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1, 1), \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
4	$p(1) > 0 \ \& \ p(-1) > 0 \ \& \ \Delta > 0 \ \& \ D > 1 \ (T < -2)$	explosive: 2 real unstable roots in $(-\infty, -1)$
5	$p(1) > 0 \ \& \ p(-1) > 0 \ \& \ \Delta < 0 \ \& \ D > 1$	explosive: complex conjugates with modulus $r = \sqrt{D} > 1$, oscillates & diverges
6	$p(1) > 0 \ \& \ p(-1) > 0 \ \& \ \Delta < 0 \ \& \ D < 1 \ (-2 < T < 2)$	stable: complex conjugates with modulus $r = \sqrt{D} < 1$, oscillates & converges
7	$\Delta > 0 \ \& \ D < 1 \ \& \ p(1) > 0 \ \& \ p(-1) > 0$	stable: 2 real roots $ b_1 < 1, b_2 < 1$
7a	$D > 0 \ \& \ T < 0$	$-1 < b_2 \leq b_1 < 0$ (same sign, and $T = b_1 + b_2$ negative) decay but can flip sign
7b	$D < 0$	$-1 < b_2 < 0 < b_1 < 1$ (opposite sign, $D = b_1 b_2$) decay but can flip sign
7c	$D > 0 \ \& \ T > 0$	$0 < b_2 \leq b_1 < 1$ (same sign, and $T = b_1 + b_2$ positive) monotone converge
8	$\Delta > 0, D > 1, T > 2$ $p(1) > 0, p(-1) > 0$	explosive: 2 real roots $b_1 > 1, b_2 > 1$

14. Blanchard and Kahn (BK) conditions

Predetermined variable: initial value is given, states like k, h, b (i.e. capital, hour, bond).

Jump variable: initial value not given, can jump to satisfy equilibrium, like c, p .

Stable roots = # predetermined variables $\Rightarrow$ Saddle path (Determinacy)

Stable roots < # predetermined variables $\Rightarrow$ Source (Explosive, no convergent equilibrium)

Stable roots > # predetermined variables $\Rightarrow$ Sink (Indeterminacy)

15. Why?

Linearize around steady state, we write deviations as a first-order system:

$$X_{t+1} = AX_t.$$

If A has eigenpairs (b_i, v_i) , the general solution can be written as sum of modes:

$$X_t = \sum_i \alpha_i b_i^t v_i.$$

If

$$|b_i| < 1,$$

mode dies out $\Rightarrow$ stable.

If

$$|b_i| > 1,$$

mode blows up $\Rightarrow$ unstable / explosive.

A convergent equilibrium path must impose: $\alpha_i = 0$ for every $|b_i| > 1$,

which is a set of restrictions, one restriction per unstable root.

Now partition variables into

{predetermined states s_t (given s_0) with dimension n_s ,

jump variables j_t (free j_0) with dimension n_j }.

$$X_t = (s_t, j_t).$$

Initial conditions pin down s_0 , but can choose j_0 .

Let

$$n_u = \# \text{ unstable roots}, \quad n_\ell = \# \text{ stable roots}.$$

$$n_u + n_\ell = n_s + n_j.$$

To satisfy n_u restrictions, we have n_j free choices to satisfy them. Thus

$$n_u = n_j \quad \Longleftrightarrow \quad n_\ell = n_s.$$

Case A

$$n_\ell = n_s \quad (n_u = n_j) :$$

jump variables jump to land on saddle path.

Case B

$$n_\ell < n_s \quad (n_u > n_j) :$$

too many unstable modes to eliminate, but too few jump variables to adjust.

Case C

$$n_\ell > n_s \quad (n_u < n_j) :$$

more jump freedom needed to kill unstable modes (multiple equilibria)